

- 6** Determine whether each function represents growth or decay:
- a** $y = 4(1.08)^x$ **b** $y = 7(0.85)^x$
- c** $y = 2e^{-0.3x}$ **d** $y = (\frac{5}{4})^x$

Compound interest

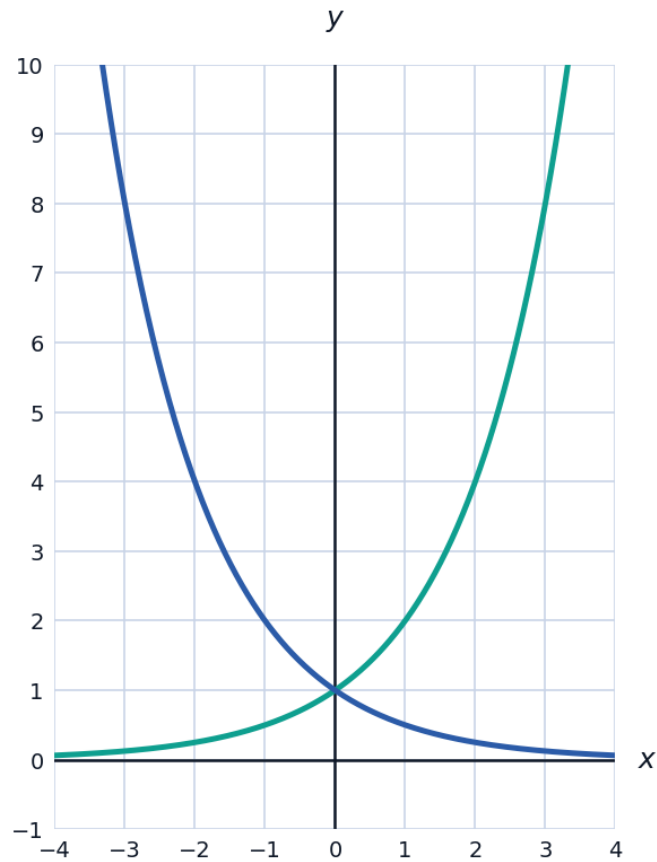
- 7 2000 riyals is invested at 5% annual interest, compounded monthly, using $A = P(1 + \frac{r}{n})^{nt}$.
- a Find the value after 3 years, to the nearest riyal.
- b Find the value after 3 years if compounded continuously instead ($A = Pe^{rt}$), to the nearest riyal.
- 8 How much must be invested now at 4% annual interest compounded annually to have 10,000 riyals in 8 years? Round to the nearest riyal.

Comparing exponential growth rates

- 9 Two investments start at 500 riyals: one grows at 4% per year, the other at 7% per year. Find the value of each after 10 years, to the nearest riyal, and their difference.
- 10 Rank the following functions from slowest to fastest growth as $x \rightarrow \infty$: $y = x^{10}$, $y = 2^x$, $y = 1.01^x$.

Comparing growth and decay graphically

- 11 The graphs of $y = 2^x$ (growth) and $y = \left(\frac{1}{2}\right)^x$ (decay) are shown together. Use the graph to describe the symmetry relating the two curves.

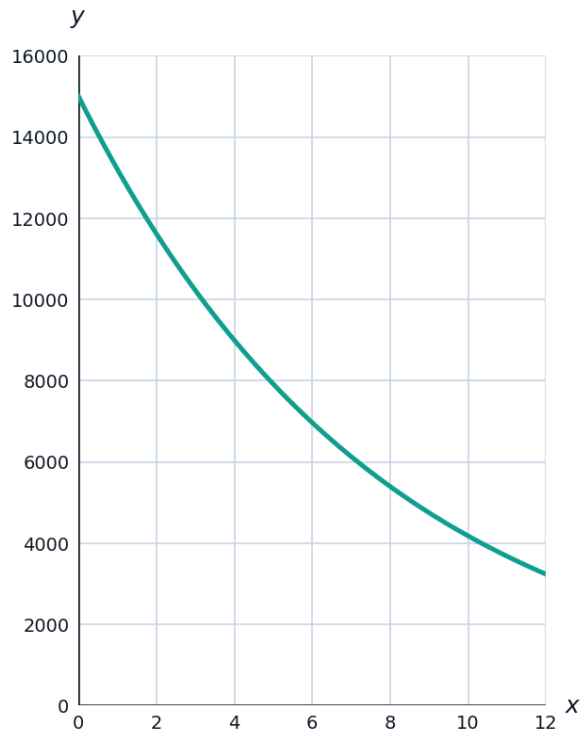


The natural exponential base e

- 12 Explain what distinguishes $e \approx 2.71828$ from other exponential bases, in terms of the slope of $y = e^x$ at $x = 0$.
- 13 Evaluate e^0 , e^1 (to 3 decimal places), and simplify $e^{\ln 5}$.

Piecewise-exponential applications

- 14 A machine depreciates in value according to $V(t) = 15000(0.88)^t$ (riyals, years). Find its value after 5 years, and the year in which its value first drops below 5000 riyals (to the nearest year).



Transformations of exponential functions

- 15 State the horizontal asymptote and range of each transformed exponential function:
- a $y = 3^x + 4$
 - b $y = -2^x$
 - c $y = 5^x - 3$
- 16 Write the equation of the function obtained by taking $y = e^x$, reflecting it across the y-axis, then shifting it down 2 units.

Comparing exponential functions numerically

- 17 For $f(x) = 3^x$, complete a short table of values at $x = -1, 0, 1, 2, 3$ and verify each value satisfies $f(x+1) = 3f(x)$.